



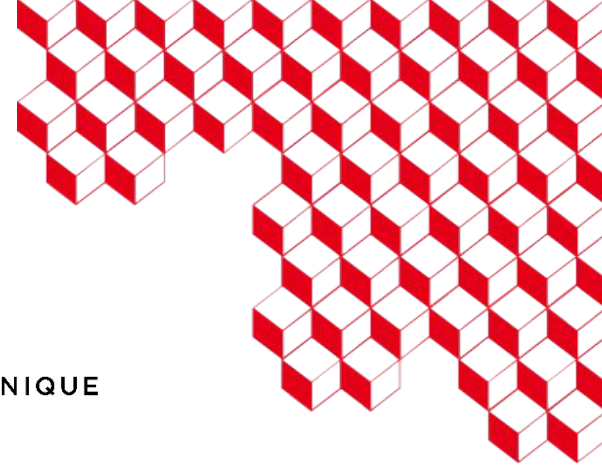
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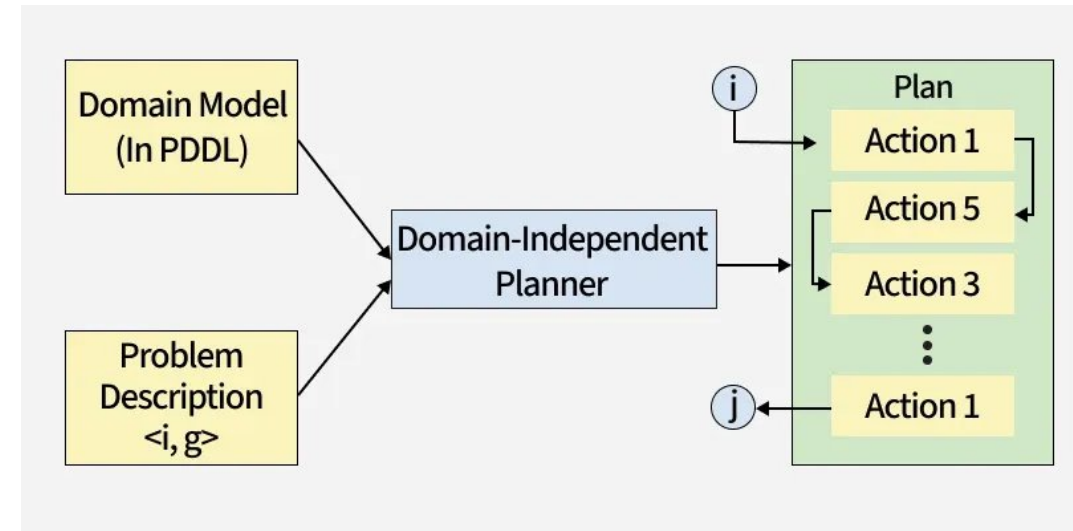
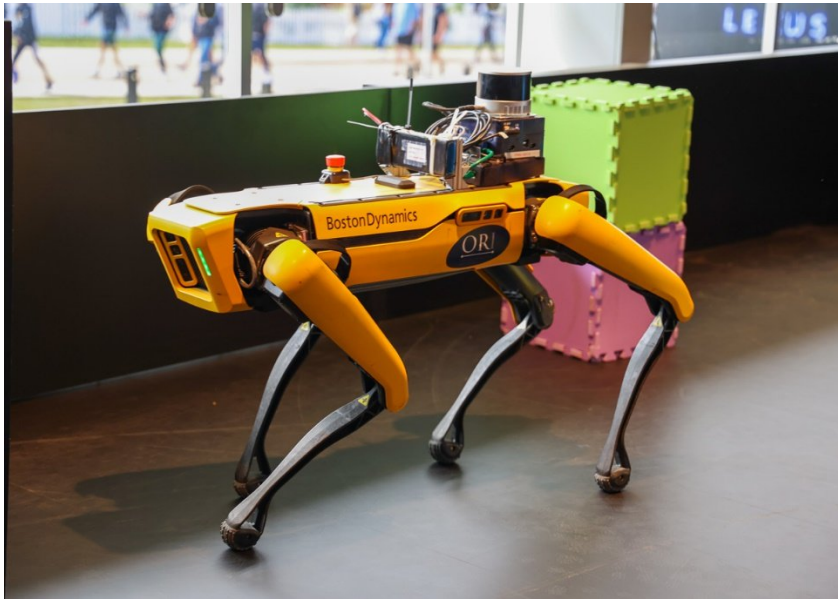
An Ontology-Based Planning Approach for Autonomous Systems

Authors : Andy Torres^{1,2}, Raphaël Lallement², Adriana Tapus¹

1 : U2IS, ENSTA Paris, Polytechnic Institute of Paris

2 : LSEA, CEA list

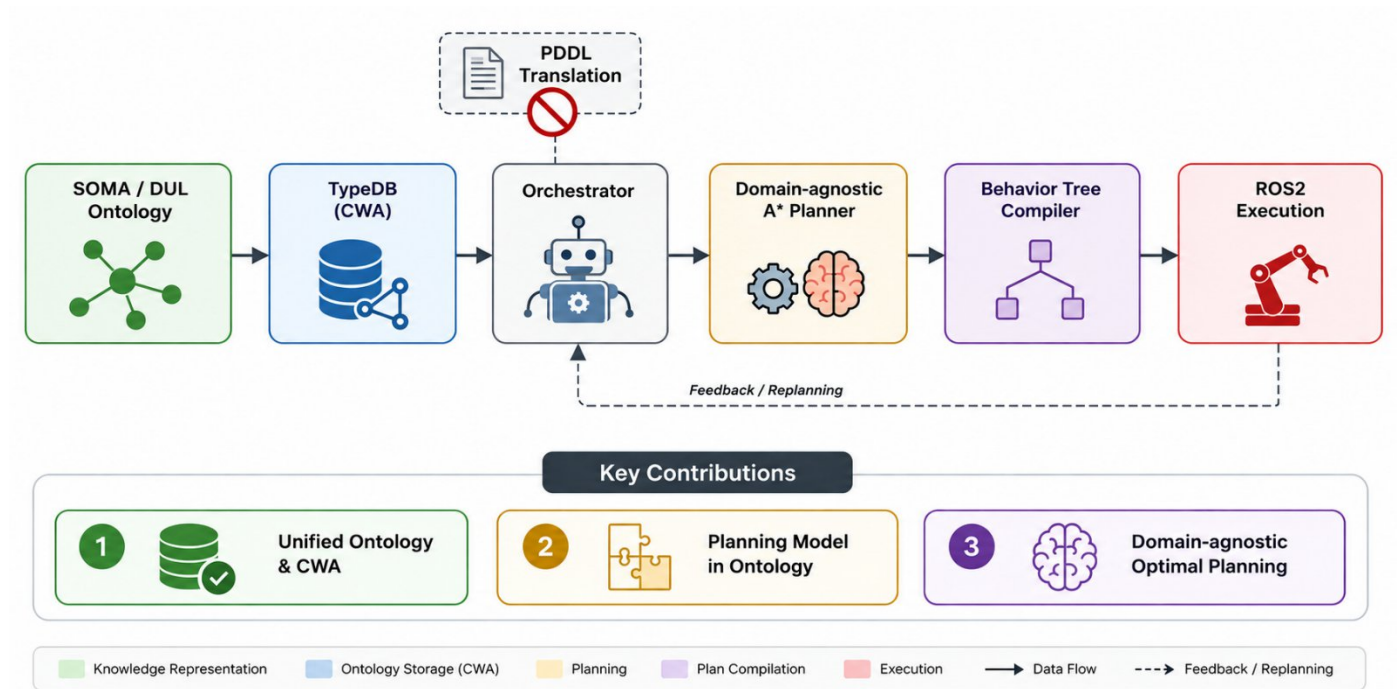
Introduction



Context and research problem

What we want to address :

- The loss of expressiveness caused by mapping rich ontological representations into PDDL
- The separation of domain knowledge from planning problem formulation
- The absence of formal constraint grounding in existing ontology-driven planners.

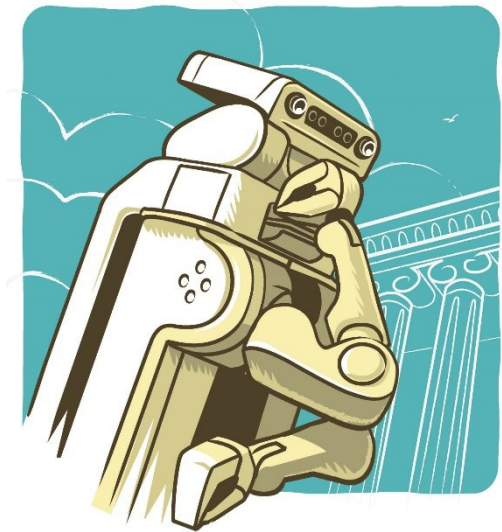


TypeDB and SOMA for Knowledge Representation

All planning knowledge is represented within a unified ontological model stored in TypeDB.

Our ontological vocabulary is grounded in the SOMA ontology. SOMA builds upon the DOLCE+DnSULtralite (DUL) upper ontology, which is itself expressed in OWL.

TypeDB's typed graph architecture natively supports n-ary relations as first-class schema constructs, allowing us to represent high level construct easily.



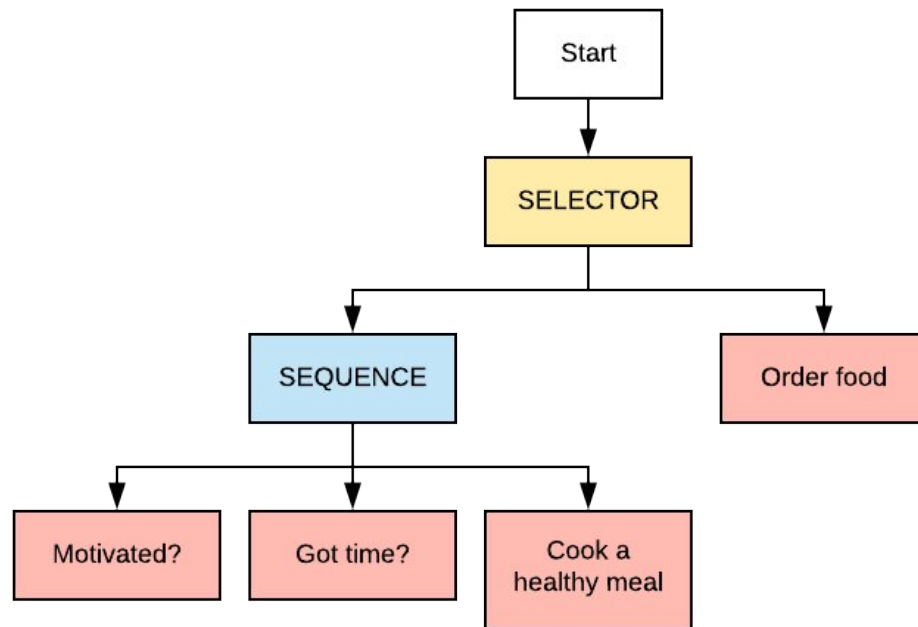
A Custom Domain-Agnostic Planner based on A*

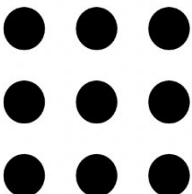
Our planner is inspired by the A* algorithm for several reasons :

- A* guarantees optimality when it comes to finding the shortest/cheapest path in a directed acyclic graph
- It has a built-in goal factor in the evaluation function ($f(n) = h(n) + g(n)$) to converge faster to the goal
- We also integrated a weight factor ($f(n) = w \cdot h(n) + g(n)$) to monitor the greediness of the search
- It's widely used in the literature and reliable



Behavior Tree Compilation and ROS2 Integration



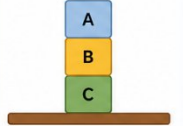
 ROS2TM

First results on classical benchmarks

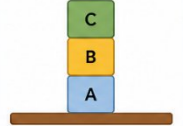
BLOCKSWORLD DOMAIN

Blocksworld consists of a set of blocks. The planner's objective is to generate a sequence of actions that transforms an initial configuration into a desired goal configuration.

INITIAL STATE
A on B on C,
C on the table

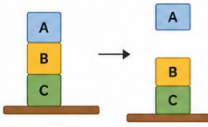


GOAL STATE
C on B and B on A,
A on the table

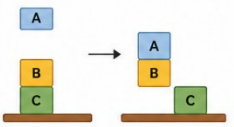


ACTIONS (examples)

1. Move block to table



2. Place block on another block

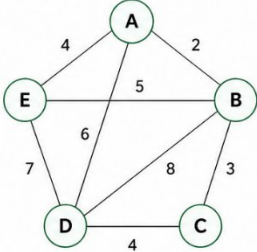


 The planner must find a sequence of actions using the available moves to transform the initial state into the goal state.

TSP DOMAIN

The TSP domain represents the Travelling Salesman Problem. Cities and the distances between them are modeled as a weighted undirected graph. The planner's goal is to find the lowest-cost route that visits each city exactly once and returns to the starting city (in our experiments, the starting point was always city A).

EXAMPLE GRAPH



GOAL
Find the lowest-cost route that:

- visits each city exactly once
- returns to the starting city (A)

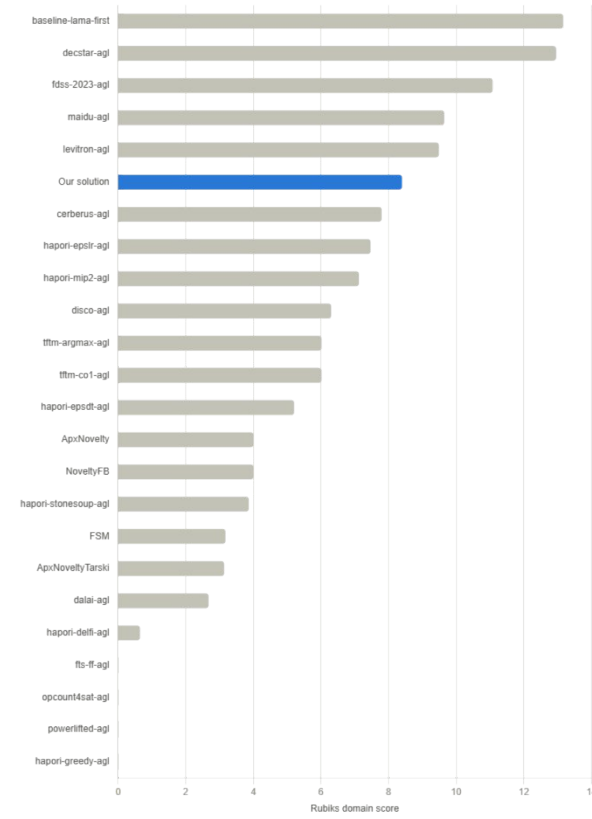
EXAMPLE SOLUTION (one of the optimal tours)

A → B → C → D → E → A = 20 (total cost)

2 2 3 4 7 4

Comparison with state-of-the-art planners

Problem	Planning Time (s)	Score
1	0.0005	1
2	0.0008	1
3	0.0010	1
4	0.0017	1
5	0.0400	1
6	0.0589	1
7	14.771	0.5279
8	16.546	0.508
9	1.0214	0.9963
10	38.1518	0.3616
11	>300	0
12	>300	0
13	>300	0
14	>300	0
15	>300	0
16	>300	0
17	>300	0
18	>300	0
19	>300	0
20	>300	0
Total Score		8.3938

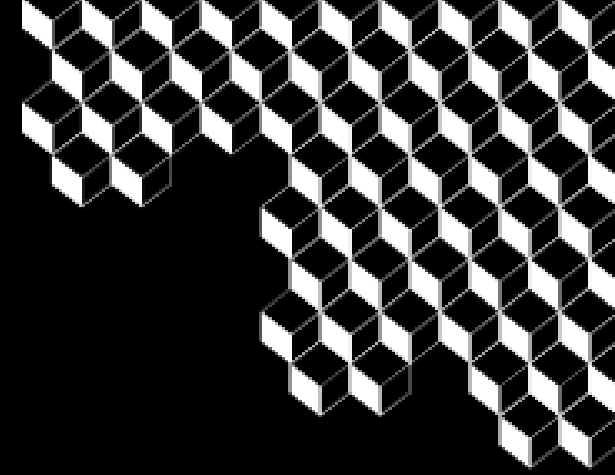


Future Work

- Add a reasoning layer inspired by OWL-DL will enable the consistency checking of the ABox/TBox during planning and a gain in expressivity to take into account complex rules involving cardinalities and disjunction for instance not inherent to TypeDB.
- Integrate the internal state (composed of the predicates that are true at the moment) even the forward-search algorithm if possible in the ontology instead of reading, extracting the planning states and create the plan externally in Python internal states.
- Model constraints in our architecture. Three categories of constraints have been identified as essential for consideration during the planning process :
 - Safety constraints in priority
 - Human interaction constraints
 - Performance constraints

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Thank you for your attention !